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Operating System

11Week [18 November - 24 November]

SOC3010 OS QUIZ5 FALL 2024

Started on	2024-11-25 18:24
State	Finished
Completed on	2024-12-01 23:58
Time taken	6 days 5 hours
Marks	1695.00/1760.00
Grade	19.26 out of 20.00 (96%)

Question 1

Incorrect

Mark 0.00 out of 10.00

..... field in inode object provides Pointers for the list of inodes of the superblock.

Answer:



The correct answer is: i_sb_list

Question 2

Correct

Mark 10.00 out of 10.00

In Ext4 file system, Journal size is

Select one:

- ☐ 1. 256MB
- ☒ 2. 128MB ✓
- ☐ 3. 64MB
- ☐ 4. 512MB

The correct answer is: 128MB

Question 3

Incorrect

Mark 0.00 out of 5.00

Full form of NTFS is

Answer:



The correct answer is: New Technology File System

Question 4

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides 128-bit filesystem identifier.

Answer:



The correct answer is: s_uuid

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Question 5

Correct

Mark 5.00 out of 5.00

State whether the following statement is TRUE or FALSE.

"In most traditional Unix-like kernels, each filesystem can be mounted only once. However, in Linux - it is possible to mount the same filesystem several times,"

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question 6

Correct

Mark 5.00 out of 5.00

The per-semaphore list is mainly used when a process invokes thefunction to force an explicit value into a primitive semaphore.

Answer: semctl()

The correct answer is: semctl()

Question 7

Correct

Mark 10.00 out of 10.00

Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.

Compute the Total number of blocks (EXACT) in the floppy disk considering its size.

Select one:

- ☐ 1. 1574
- ☒ 2. 1474 ✓
- ☐ 3. 1400
- ☐ 4. 1500

The correct answer is: 1474

Question 8

Correct

Mark 5.00 out of 5.00

The following command displays a filesystem - both superblock and all of the block group descriptor detail information.

Select one:

- ☐ 1. \$dumpe2fs -h /dev/sda1
- ☒ 2. \$dumpe2fs -f /dev/sda1 ✓
- ☐ 3. \$dumpe2fs -x /dev/sda1
- ☐ 4. \$dumpe2fs -b /dev/sda1

The correct answer is: \$dumpe2fs -f /dev/sda1

Question **9**

Incorrect

Mark 0.00 out of 5.00

In ext2_inode structure, field provides number of data blocks of the file in terms of sectors.

Select one:

- ☒ 1. i_block ✖
- ☐ 2. i_size
- ☐ 3. i_file_acl
- ☐ 4. i_blocks

The correct answer is: i_blocks

Question **10**

Correct

Mark 5.00 out of 5.00

Assume a data block size of 4KB, find the file block number for the given file offset 14320.

Select one:

- ☐ 1. 5
- ☐ 2. 6
- ☒ 3. 3 ✔
- ☐ 4. 4

The correct answer is: 3

Question **11**

Correct

Mark 10.00 out of 10.00

..... field in inode object provides the inode number

Answer: ✔

The correct answer is: i_ino

Question **12**

Incorrect

Mark 0.00 out of 5.00

..... avoids the time-consuming check that is automatically performed on a filesystem when it is abruptly unmounted for instance, as a consequence of a system crash.

Answer: ✖

The correct answer is: Journaling

Question **13**

Correct

Mark 5.00 out of 5.00

Files larger than must be opened on 32-bit architectures by specifying the O_LARGEFILE opening flag.

Select one:

- ☐ 1. 1GB
- ☒ 2. 2 GB ✓
- ☐ 3. 512KB
- ☐ 4. 256KB

The correct answer is: 2 GB

Question **14**

Correct

Mark 5.00 out of 5.00

.....field in msg_queue data structure provides number of messages in queue

Select one:

- ☒ 1. q_qnum ✓
- ☐ 2. q_messages
- ☐ 3. q_qbytes
- ☐ 4. q_qbytes

The correct answer is: q_qnum

Question **15**

Correct

Mark 5.00 out of 5.00

A Page Fault occurs when a process tries to access a location of an IPC shared memory region whose underlying page frame has not been assigned. The corresponding exception handler determines that the faulty address is inside the process address space and that the corresponding Page Table entry is null; therefore, it invokes the function.

Answer: ✓

The correct answer is: do_no_page()

Question **16**

Correct

Mark 5.00 out of 5.00

The Symbolic link file is associated with a file type number

Answer: ✓

The correct answer is: 7

Question **17**

Correct

Mark 5.00 out of 5.00

The field in the sem_array data structure points to an array of sem data structures, one for every IPC primitive semaphore.

Select one:

- ☐ 1. status
- ☐ 2. sops
- ☐ 3. sma
- ☒ 4. sem_base ✓

The correct answer is: sem_base

Question **18**

Correct

Mark 5.00 out of 5.00

Write a single UNIX command to replace the following two commands :

\$ ls > temp

\$ more < temp

Answer: ✓

The correct answer is: ls | more

Question **19**

Correct

Mark 5.00 out of 5.00

Space management must be ; that is, the kernel should be able to quickly derive from a file offset the corresponding logical block number in the Ext2 partition.

Answer: ✓

The correct answer is: time-efficient

Question **20**

Correct

Mark 5.00 out of 5.00

In a pipe_inode_info structure, field provides a pointer to Pipe/FIFO wait queue

Answer: ✓

The correct answer is: wait

Question **21**

Correct

Mark 5.00 out of 5.00

In Ext2 Block group descriptor structure, field provides Number of free inodes in the group

Answer: ✓

The correct answer is: bg_free_inodes_count

Question **22**
Correct
Mark 10.00 out of 10.00

Assume a data block size of 1KB, find the file block number for the given file offset 14320.

Select one:

- ☐ 1. 19
- ☐ 2. 16
- ☐ 3. 15
- ☒ 4. 13 ✓

The correct answer is: 13

Question **23**
Correct
Mark 10.00 out of 10.00

If disk block size = 16KB, # of inodes = 32768 and Partition size = 512GB then determine the Number of Blocks in a blockgroup in the file system.

Select one:

- ☒ 1. 131072 ✓
- ☐ 2. 16384
- ☐ 3. 32786
- ☐ 4. 65536

The correct answer is: 131072

Question **24**
Correct
Mark 10.00 out of 10.00

For a given 32-GB Ext2 partition with a 4-KB block size, the approximate number of Block Groups in the disk partition is

Select one:

- ☒ 1. 256 ✓
- ☐ 2. 4096
- ☐ 3. 8192
- ☐ 4. 1024

The correct answer is: 256

Question **25**
Correct
Mark 5.00 out of 5.00

Consider Data Block addressing in Ext2 disk file system with a block size of b bytes.

Determine the range of file block numbers of data blocks (i.e., number or index of the data blocks) that can be addressed by first 12 components i.e., $i_block[0]$ to $i_block[11]$ of the i_block array

Select one:

- ☐ 1. 12 to $(b/4) + 11$
- ☒ 2. 0 to 11 ✓
- ☐ 3. $(b/4)^2 + (b/4) + 12$ to $(b/4)^3 + (b/4)^2 + (b/4) + 11$
- ☐ 4. $(b/4) + 12$ to $(b/4)^2 + (b/4) + 11$

The correct answer is: 0 to 11

Question **26**

Incorrect

Mark 0.00 out of 5.00

Theincludes all sem_undo data structures corresponding to IPC semaphores on which the process has performed undoable operations.

Answer: per-semaphore list



The correct answer is: per-process list

Question **27**

Correct

Mark 20.00 out of 20.00

Assume a data block size of 1KB, if the file block number is 6623 then compute the corresponding logical block number of the data block.

Select one:

- ☐ 1. i_block[14], first level indirectblock[0] & second level indirectblock0[25]
- ☐ 2. i_block[12] & first level indirectblock[255]
- ☒ 3. i_block[13], first level indirectblock[0] & second level indirectblock24[211] ✓
- ☐ 4. i_block[13], first level indirectblock[1] & second level indirectblock25[211]

The correct answer is: i_block[13], first level indirectblock[0] & second level indirectblock24[211]

Question **28**

Correct

Mark 5.00 out of 5.00

..... function is invoked to create IPC Shared memory resource.

Select one:

- ☐ 1. shmat()
- ☐ 2. msgget()
- ☐ 3. semget()
- ☒ 4. shmget() ✓

The correct answer is: shmget()

Question **29**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides number of inodes per block group

Answer: s_inodes_per_group



The correct answer is: s_inodes_per_group

Question **30**

Correct

Mark 5.00 out of 5.00

When the kernel has to locate a block holding data for an Ext2 regular file, it invokes the function. If the block does not exist, the function automatically allocates the block to the file.

Select one:

- ☐ 1. ext2_new_inode()
- ☐ 2. ext2_free_blocks()
- ☒ 3. ex2_get_block() ✓
- ☐ 4. ext2_free_inode()

The correct answer is: ex2_get_block()

Question **31**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides number of reserved blocks

Select one:

- ☐ 1. s_block_group_nr
- ☐ 2. s_blocks_count
- ☒ 3. s_r_blocks_count ✓
- ☐ 4. s_blocks_per_group

The correct answer is: s_r_blocks_count

Question **32**

Correct

Mark 5.00 out of 5.00

" Memory regions used for IPC shared memory always define the nopage method. "

Say True or False

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **33**

Correct

Mark 5.00 out of 5.00

When two or more processes wish to communicate through an IPC resource, they all refer to the IPC identifier of the resource.

Say True or False

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **34**

Correct

Mark 10.00 out of 10.00

For a given 512-GB Ext4 partition with a 8-KB block size, the approximate number of Block Groups in the disk partition is

Select one:

- ☐ 1. 256
- ☐ 2. 512
- ☐ 3. 4096
- ☒ 4. 1024 ✓

The correct answer is: 1024

Question **35**

Correct

Mark 5.00 out of 5.00

The field of an inode points to the block containing the extended attributes.

Answer: ✓

The correct answer is: i_file_acl

Question **36**

Correct

Mark 5.00 out of 5.00

The Unknown file is associated with a file type number

Answer: ✓

The correct answer is: 0

Question **37**

Correct

Mark 10.00 out of 10.00

Suppose that each block group contains 4,096 inodes, the address on disk of inode 26,042 can be given as

Select one:

- ☐ 1. Blockgroup 6, 13021
- ☒ 2. Blockgroup 6, 1466 ✓
- ☐ 3. Blockgroup 7, 26042
- ☐ 4. Blockgroup 7, 1467

The correct answer is: Blockgroup 6, 1466

Question **38**

Correct

Mark 5.00 out of 5.00

The counters inside an IPC semaphore are referred to as

Answer: ✓

The correct answer is: primitive semaphores

Question **39**
Correct
Mark 20.00 out of 20.00

In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 and 512 Blockgroups, then the number of data blocks available for storing file data is blocks

Select one:

- ☒ 1. 7921 ✓
- ☐ 2. 1024
- ☐ 3. 4355
- ☐ 4. 8192

The correct answer is: 7921

Question **40**
Correct
Mark 5.00 out of 5.00

In ext4, by default, a filesystem is allowed to increase in size by a factor ofx over the original filesystem size.

Answer: ✓

The correct answer is: 1024

Question **41**
Correct
Mark 10.00 out of 10.00

For a given Ext4 file system having 8192 inodes in each blockgroup with blocksize 4KB, the total number of blocks required by the Inode Table is

Select one:

- ☐ 1. 128
- ☐ 2. 1024
- ☒ 3. 512 ✓
- ☐ 4. 256

The correct answer is: 512

Question **42**
Correct
Mark 10.00 out of 10.00

In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the Size of one blockgroup.

Select one:

- ☐ 1. 256MB
- ☒ 2. 512MB ✓
- ☐ 3. 128MB
- ☐ 4. 1024MB

The correct answer is: 512MB

Question **43**

Correct

Mark 10.00 out of 10.00

..... field in superblock object provides list of inodes waiting to be written to disk

Answer: ✓

The correct answer is: s_io

Question **44**

Correct

Mark 5.00 out of 5.00

..... command is used to perform automatic consistency checks on the filesystem status

Select one:

- ☐ 1. mkfs
- ☒ 2. e2fsck ✓
- ☐ 3. tune2fs
- ☐ 4. dumpe2fs

The correct answer is: e2fsck

Question **45**

Correct

Mark 5.00 out of 5.00

..... command is used to create a Ext2 filesystem on a hard disk

Select one:

- ☒ 1. mke2fs ✓
- ☐ 2. e2fsck
- ☐ 3. tune2fs
- ☐ 4. dumpe2fs

The correct answer is: mke2fs

Question **46**

Correct

Mark 5.00 out of 5.00

The number of blocks required to store Inode Bitmap in a Block group is

Answer: ✓

The correct answer is: 1

Question **47**

Correct

Mark 5.00 out of 5.00

All fields in Ext4 are written to disk in

Select one:

- ☐ 1. only unsigned form
- ☐ 2. only signed form
- ☐ 3. big-endian order
- ☒ 4. little-endian order ✓

The correct answer is: little-endian order

Question **48**

Correct

Mark 5.00 out of 5.00

In Unix System V, Maximum total size of the messages in a queue by default isbytes

Answer: 16384



The correct answer is: 16384

Question **49**

Correct

Mark 5.00 out of 5.00

In Ext2 Block group descriptor structure, field provides Number of directories in the group

Answer: bg_used_dirs_count



The correct answer is: bg_used_dirs_count

Question **50**

Correct

Mark 5.00 out of 5.00

The function is invoked to "attach" an IPC shared memory region to a process.

Answer: shmat()



The correct answer is: shmat()

Question **51**

Correct

Mark 5.00 out of 5.00

.....are counters used to provide controlled access to shared data structures for multiple processes.

Select one:

- ☐ 1. IPC Shared Memory
- ☒ 2. IPC semaphores ✓
- ☐ 3. IPC Message Queues
- ☐ 4. Kernel Semaphore

The correct answer is: IPC semaphores

Question **52**

Correct

Mark 5.00 out of 5.00

"The goal of a journaling filesystem is to avoid running time-consuming consistency checks on the whole filesystem by looking instead in a special disk area that contains the most recent disk write operations named journal."

Say TRUE OR FALSE

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **53**

Correct

Mark 5.00 out of 5.00

When a process "attaches" a segment, the kernel invokes and creates a new shared memory mapping of the file in the address space of the process.

Answer: ✓

The correct answer is: do_mmap()

Question **54**

Correct

Mark 5.00 out of 5.00

Thefunction is invoked to "detach" an IPC shared memory region specified by its IPC identifier that is, to remove the corresponding memory region from the process's address space.

Answer: ✓

The correct answer is: shmdt()

Question **55**

Correct

Mark 5.00 out of 5.00

In order to compute the IPC Identifier using the expression $\text{IPC Identifier} = s * M + i$, Linux uses the value of M set to, where s starts from 0 and $0 \leq i < M$.

Select one:

- ☐ 1. 65536
- ☒ 2. 32768 ✓
- ☐ 3. 16384
- ☐ 4. 32767

The correct answer is: 32768

Question **56**

Correct

Mark 10.00 out of 10.00

Assume a data block size of 4KB, find the file block number for the given file offset 6782341.

Select one:

- ☐ 1. 1755
- ☒ 2. 1655 ✓
- ☐ 3. 1855
- ☐ 4. 1555

The correct answer is: 1655

Question **57**

Correct

Mark 5.00 out of 5.00

A process using IPC message queue resource invokes the function to receive a message.

Answer: ✓

The correct answer is: msgrcv()

Question **58**

Correct

Mark 10.00 out of 10.00

Suppose that each block group contains 8,192 inodes, the address on disk of inode 23,146 can be given as

Select one:

- ☐ 1. Blockgroup 1, 6762
- ☐ 2. Blockgroup 3, 6762
- ☒ 3. Blockgroup 2, 6762 ✓
- ☐ 4. Blockgroup 2, 6761

The correct answer is: Blockgroup 2, 6762

Question **59**

Correct

Mark 10.00 out of 10.00

In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 and 512 Blockgroups, then the size of one blockgroup is

Select one:

- ☐ 1. 32MB
- ☒ 2. 8MB ✓
- ☐ 3. 64MB
- ☐ 4. 8GB

The correct answer is: 8MB

Question **60**

Correct

Mark 10.00 out of 10.00

Assume a data block size of 4KB, if the file block number is 3 then compute the corresponding logical block number of the data block.

Select one:

- ☒ 1. i_block[3] ✓
- ☐ 2. i_block[12] & first level indirectblock[2]
- ☐ 3. i_block[13] & first level indirectblock[3]
- ☐ 4. i_block[12] & first level indirectblock[3]

The correct answer is: i_block[3]

Question **61**

Correct

Mark 5.00 out of 5.00

In a pipe_inode_info structure, field provides Array of pipe's buffer descriptors.

Answer:

The correct answer is: buf

Question **62**

Correct

Mark 5.00 out of 5.00

In ext2 i_node structure, field provides Hard links counter.

Answer:



The correct answer is: i_links_count

Question **63**

Correct

Mark 10.00 out of 10.00

Consider Data Block addressing in Ext2/ext4 disk file system with a block size of 4KB.

Determine the maximum size of the file that can be addressed by 12th component of the i_block array.

Select one:

- ☐ 1. 4TB
- ☐ 2. 48KB
- ☐ 3. 4GB
- ☒ 4. 4MB ✓

The correct answer is: 4MB

Question **64**

Correct

Mark 5.00 out of 5.00

The number of bytes currently written in all pipe buffers and yet to be read is called

Answer:



The correct answer is: pipe size

Question **65**

Correct

Mark 5.00 out of 5.00

When shell executes the following command, process takes no inputs from the pipes

`$ head -60 tfile | tail -1 | wc -c`

Select one:

- ☒ 1. head -60 tfile ✓
- ☐ 2. wc -c
- ☐ 3. tail -1
- ☐ 4. shell

The correct answer is: head -60 tfile

Question **66**

Correct

Mark 10.00 out of 10.00

If disk block size = 16KB, # of inodes = 32768 and Partition size = 512GB then determine the Size of the inode table in blocks in each of the block group in the Ext4 file system.

Select one:

- ☐ 1. 256
- ☒ 2. 512 ✓
- ☐ 3. 1024
- ☐ 4. 2048

The correct answer is: 512

Question **67**

Correct

Mark 20.00 out of 20.00

Consider Data Block addressing in Ext2/ext4 disk file system with a block size of 4KB.

Determine the maximum size of the file that can be addressed by 14th component of the i_block array.

Select one:

- ☒ 1. 4TB ✓
- ☐ 2. 48KB
- ☐ 3. 4GB
- ☐ 4. 4MB

The correct answer is: 4TB

Question **68**

Correct

Mark 5.00 out of 5.00

The message text starts right after the msg_msg descriptor; if the message is longer thanbytes (the page size minus the size of the msg_msg descriptor), it continues on another page, whose address is stored in the next field of the msg_msg descriptor.

Select one:

- ☒ 1. 4072 ✓
- ☐ 2. 4096
- ☐ 3. 3072
- ☐ 4. 4000

The correct answer is: 4072

Question **69**

Correct

Mark 5.00 out of 5.00

When shell executes the following command, process writes the output to the display(screen)

```
$ head -60 tfile | tail -1 | wc -c
```

Select one:

- ☐ 1. tail -1
- ☐ 2. shell
- ☒ 3. wc -c ✓
- ☐ 4. head -60 tfile

The correct answer is: wc -c

Question **70**

Correct

Mark 5.00 out of 5.00

When the e2fsck program executes a consistency check on the filesystem status, it refers to the superblock and the block group descriptors stored in block group -----, and then copies them into all other block groups.

Select one:

- ☒ 1. 0 ✓
- ☐ 2. 5
- ☐ 3. 7
- ☐ 4. 3

The correct answer is: 0

Question **71**

Correct

Mark 5.00 out of 5.00

Thefunction, which is invoked by do_exit(), walks through the list and reverses the effect of any unbalanced operation for every IPC semaphore touched by the process.

Answer: ✓

The correct answer is: exit_sem()

Question **72**

Correct

Mark 5.00 out of 5.00

The following command displays the detailed group information block numbers in hexadecimal format.

Select one:

- ☐ 1. \$dumpe2fs -b /dev/sda1
- ☐ 2. \$dumpe2fs -h /dev/sda1
- ☐ 3. \$dumpe2fs -f /dev/sda1
- ☒ 4. \$dumpe2fs -x /dev/sda1 ✓

The correct answer is: \$dumpe2fs -x /dev/sda1

Question **73**

Correct

Mark 5.00 out of 5.00

The Directory file is associated with a file type number

Answer:

2



The correct answer is: 2

Question **74**

Correct

Mark 5.00 out of 5.00

....., Allow processes to exchange information of large amounts of data

Select one:

- ☐ 1. Semaphore
- ☐ 2. Sockets
- ☐ 3. Message Queues
- ☒ 4. Shared Memory ✓

The correct answer is: Shared Memory

Question **75**

Incorrect

Mark 0.00 out of 5.00

In ext2_inode structure, field provides pointers to data blocks.

Select one:

- ☒ 1. i_blocks ✗
- ☐ 2. i_file_acl
- ☐ 3. i_block
- ☐ 4. i_size

The correct answer is: i_block

Question **76**

Correct

Mark 10.00 out of 10.00

Consider Data Block addressing in Ext2 disk file system with a block size of b bytes.

Determine the range of file block numbers of data blocks (i.e., number or index of the data blocks) that can be addressed by 12th component $i_block[12]$ of the i_block array

Select one:

- ☐ 1. 0 to 11
- ☐ 2. $(b/4)^2 + (b/4) + 12$ to $(b/4)^3 + (b/4)^2 + (b/4) + 11$
- ☒ 3. 12 to $(b/4) + 11$ ✓
- ☐ 4. $(b/4) + 12$ to $(b/4)^2 + (b/4) + 11$

The correct answer is: 12 to $(b/4) + 11$

Question **77**

Correct

Mark 5.00 out of 5.00

In Ext2 Block group descriptor structure, field provides Block number of inode bitmap.

Answer: ✓

The correct answer is: bg_inode_bitmap

Question **78**

Correct

Mark 5.00 out of 5.00

..... field in shmid_kernel data structure provides segment size in bytes

Select one:

- ☒ 1. shm_segsz ✓
- ☐ 2. shm_atim
- ☐ 3. shm_file
- ☐ 4. shm_nattch

The correct answer is: shm_segsz

Question **79**

Correct

Mark 15.00 out of 15.00

Consider Data Block addressing in Ext2 disk file system with a block size of b bytes.

Determine the range of file block numbers of data blocks (i.e., number or index of the data blocks) that can be addressed by 13th component `i_block[13]` of the `i_block` array

Select one:

- ☐ 1. $(b/4)^2 + (b/4) + 12$ to $(b/4)^3 + (b/4)^2 + (b/4) + 11$
- ☐ 2. 0 to 11
- ☐ 3. 12 to $(b/4) + 11$
- ☒ 4. $(b/4) + 12$ to $(b/4)^2 + (b/4) + 11$ ✓

The correct answer is: $(b/4) + 12$ to $(b/4)^2 + (b/4) + 11$

Question **80**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides filesystem size in blocks.

Select one:

- ☒ 1. s_blocks_count ✓
- ☐ 2. s_free_blocks_count
- ☐ 3. s_log_block_size
- ☐ 4. s_blocks_per_group

The correct answer is: s_blocks_count

Question **81**

Correct

Mark 5.00 out of 5.00

In Ext2 Block group descriptor structure, field provides Number of free blocks in the group.

Answer: ✓

The correct answer is: bg_free_blocks_count

Question **82**

Incorrect

Mark 0.00 out of 5.00

In Ext2 superblock structure, field provides number of blocks per block group

Answer: ✗

The correct answer is: s_blocks_per_group

Question **83**

Correct

Mark 5.00 out of 5.00

If the symbolic link represents a short pathname with at most characters, it can be stored in the inode and can thus be translated without reading a data block.

Select one:

- ☐ 1. 128
- ☐ 2. 256
- ☐ 3. 128
- ☒ 4. 60 ✓

The correct answer is: 60

Question **84**

Correct

Mark 5.00 out of 5.00

A pipe is implemented as a set of VFS objects, which have no corresponding disk images.

Say True or False

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **85**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides number of blocks to preallocate for files

Answer: ✓

The correct answer is: s_prealloc_blocks

Question **86**

Correct

Mark 5.00 out of 5.00

No data blocks are required for one of these kinds of files.

Select one:

- ☒ 1. Device file ✓
- ☐ 2. Symbolic Link
- ☐ 3. Regular file
- ☐ 4. Directory

The correct answer is: Device file

Question **87**

Correct

Mark 20.00 out of 20.00

Consider Data Block addressing in Ext2 disk file system with a block size of b bytes.

Determine the range of file block numbers of data blocks (i.e., number or index of the data blocks) that can be addressed by 14th component $i_block[14]$ of the i_block array

Select one:

- ☐ 1. 0 to 11
- ☒ 2. $(b/4)^2 + (b/4) + 12$ to $(b/4)^3 + (b/4)^2 + (b/4) + 11$ ✓
- ☐ 3. 12 to $(b/4) + 11$
- ☐ 4. $(b/4) + 12$ to $(b/4)^2 + (b/4) + 11$

The correct answer is: $(b/4)^2 + (b/4) + 12$ to $(b/4)^3 + (b/4)^2 + (b/4) + 11$

Question **88**

Correct

Mark 10.00 out of 10.00

Given Ex2 filesystem with default data block size=4KB, write the file size and total number of blocks assigned to a file hole when the following command is executed.

```
$ echo -n "XYZ" | dd of=/tmp/hole bs=4096 seek=7
```

Select one:

- ☐ 1. File Size = 3, block required=1
- ☐ 2. File Size = 28675, block required=8
- ☐ 3. File Size = 28672, block required=7
- ☒ 4. File Size = 28675, block required=1 ✓

The correct answer is: File Size = 28675, block required=1

Question **89**

Correct

Mark 5.00 out of 5.00

In Unix System V, Number of IPC message queue resources allowed by default is

Answer: ✓

The correct answer is: 16

Question **90**

Correct

Mark 10.00 out of 10.00

For a given 64GB partition having Ext2 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the size of one blockgroup is blocks

Select one:

- ☒ 1. 128MB ✓
- ☐ 2. 2048MB
- ☐ 3. 512MB
- ☐ 4. 1024MB

The correct answer is: 128MB

Question **91**

Correct

Mark 5.00 out of 5.00

A process using IPC message queue resource invokes the function to send a message.

Answer: ✓

The correct answer is: msgsnd()

Question **92**

Correct

Mark 5.00 out of 5.00

In a pipe_inode_info structure, field provides Number of buffers containing data to be read.

Answer: ✓

The correct answer is: nrbufs

Question **93**

Correct

Mark 5.00 out of 5.00

"To balance the number of regular files and directories in a block group, Ext2 introduces a debt parameter for every block group. "

SAY TRUE OR FALSE

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **94**

Correct

Mark 20.00 out of 20.00

.....fields in the Ext2 file system data structures are cached only in Dynamic memory.

Select one or more:

- ☐ 1. Free block
- ☒ 2. Data block ✓
- ☐ 3. Superblock
- ☒ 4. Block Bitmap ✓
- ☐ 5. Free inode
- ☒ 6. inode ✓
- ☒ 7. Inode Bitmap ✓
- ☐ 8. Block Group Descriptors

The correct answers are: Block Bitmap , Inode Bitmap, inode, Data block

Question **95**

Correct

Mark 10.00 out of 10.00

For a given 512-GB Ext4 partition with a 4-KB block size, the approximate number of Block Groups in the disk partition is

Select one:

- ☐ 1. 2048
- ☐ 2. 512
- ☒ 3. 4096 ✓
- ☐ 4. 1024

The correct answer is: 4096

Question **96**

Correct

Mark 5.00 out of 5.00

In Ext2 Block group descriptor structure, field provides Block number of first inode table block.

Answer:

The correct answer is: bg_inode_table

Question **97**

Correct

Mark 5.00 out of 5.00

Consider Data Block addressing in Ext2 disk file system with a block size of 4KB.

Determine the maximum size of the file that can be addressed by first 12 components of the i_block array.

Select one:

- ☐ 1. 4MB
- ☒ 2. 48KB ✓
- ☐ 3. 4GB
- ☐ 4. 4TB

The correct answer is: 48KB

Question **98**

Correct

Mark 5.00 out of 5.00

----- field in msg_queue data structure provides maximum number of bytes in queue

Select one:

- ☒ 1. q_qbytes ✓
- ☐ 2. q_messages
- ☐ 3. q_qnum
- ☐ 4. q_qcbytes

The correct answer is: q_qbytes

Question **99**

Correct

Mark 5.00 out of 5.00

The following command only displays the superblock information and not any of the block group descriptor detail information.

Select one:

- ☐ 1. \$dumpe2fs -x /dev/sda1
- ☐ 2. \$dumpe2fs -b /dev/sda1
- ☐ 3. \$dumpe2fs -f /dev/sda1
- ☒ 4. \$dumpe2fs -h /dev/sda1 ✓

The correct answer is: \$dumpe2fs -h /dev/sda1

Question **100**

Correct

Mark 10.00 out of 10.00

Many Unix systems provide, besides the pipe() system call, two wrapper functions named & that handle all the work usually done when using pipes.

Select one or more:

- ☐ 1. pipe()
- ☐ 2. close()
- ☒ 3. pclose() ✓
- ☒ 4. popen() ✓

The correct answers are: popen(), pclose()

Question **101**

Correct

Mark 5.00 out of 5.00

The size of each inode in the inode table in ext4 isbytes

Select one:

- ☒ 1. 256 ✓
- ☐ 2. 32
- ☐ 3. 64
- ☐ 4. 128

The correct answer is: 256

Question **102**

Correct

Mark 10.00 out of 10.00

Assume a data block size of 1KB, find the file block number for the given file offset 6782341.

Select one:

- ☐ 1. 6723
- ☒ 2. 6623 ✓
- ☐ 3. 6523
- ☐ 4. 6423

The correct answer is: 6623

Question **103**

Correct

Mark 5.00 out of 5.00

In ext2_inode structure, field provides file type and access rights

Select one:

- ☒ 1. i_mode ✓
- ☐ 2. i_blocks
- ☐ 3. i_size
- ☐ 4. i_file_acl

The correct answer is: i_mode

Question **104**

Correct

Mark 5.00 out of 5.00

When the inode object refers to a pipe, its field points to a pipe_inode_info structure.

Answer: ✓

The correct answer is: i_pipe

Question **105**

Correct

Mark 5.00 out of 5.00

All fields in jbd2 (the journal) are written to disk in

Select one:

- ☐ 1. only signed form
- ☒ 2. big-endian order ✓
- ☐ 3. only unsigned form
- ☐ 4. little-endian order

The correct answer is: big-endian order

Question **106**

Correct

Mark 10.00 out of 10.00

In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the number of Data blocks available for storing File contents in each Blockgroup.

Select one:

- ☒ 1. 63997 ✓
- ☐ 2. 32153
- ☐ 3. 64998
- ☐ 4. 65018

The correct answer is: 63997

Question **107**

Correct

Mark 5.00 out of 5.00

In a pipe_inode_info structure, field provides Index of first buffer containing data to be read.

Answer: ✓

The correct answer is: curbuf

Question **108**

Correct

Mark 10.00 out of 10.00

In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the Number of blockgroups

Select one:

- ☐ 1. 1024
- ☐ 2. 128
- ☐ 3. 256
- ☒ 4. 512 ✓

The correct answer is: 512

Question **109**

Correct

Mark 10.00 out of 10.00

For a given Ext2 file system having 8192 inodes in each blockgroup with blocksize 4KB, the total number of blocks required by the Inode Table is

Select one:

- ☐ 1. 512
- ☐ 2. 128
- ☒ 3. 256 ✓
- ☐ 4. 1024

The correct answer is: 256

Question **110**

Correct

Mark 10.00 out of 10.00

For given a 64GB partition having Ext4 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the number of data blocks available for storing file data is blocks

Select one:

- ☒ 1. 31229 ✓
- ☐ 2. 30009
- ☐ 3. 30229
- ☐ 4. 32768

The correct answer is: 31229

Question **111**

Correct

Mark 5.00 out of 5.00

..... field in shmid_kernel data structure provides a pointer to the special file of the segment

Select one:

- ☐ 1. shm_atim
- ☒ 2. shm_file ✓
- ☐ 3. shm_segsz
- ☐ 4. shm_nattch

The correct answer is: shm_file

Question **112**

Correct

Mark 10.00 out of 10.00

Given Ex2 filesystem with default data block size=4KB, write the file size in bytes and total number of blocks assigned to a file hole when the following command is executed.

```
$ echo -n "XYZ" | dd of=/tmp/hole bs=1024 seek=4
```

Select one:

- ☐ 1. File Size = 4096, block required=5
- ☐ 2. File Size = 4099, block required=5
- ☒ 3. File Size = 4099, block required=1 ✓
- ☐ 4. File Size = 3, block required=1

The correct answer is: File Size = 4099, block required=1

Question **113**

Correct

Mark 10.00 out of 10.00

.....fields in the Ext2 file system data structures are never cached.

Select one or more:

- ☒ 1. Free inode ✓
- ☐ 2. Superblock
- ☐ 3. Inode Bitmap
- ☐ 4. Block Bitmap
- ☒ 5. Free block ✓

The correct answers are: Free inode, Free block

Question **114**

Correct

Mark 10.00 out of 10.00

For a given 32-GB Ext2 partition with a 1-KB block size, the approximate number of Block Groups in the disk partition is

Select one:

- ☐ 1. 256
- ☐ 2. 8192
- ☒ 3. 4096 ✓
- ☐ 4. 1024

The correct answer is: 4096

Question **115**

Incorrect

Mark 0.00 out of 5.00

.....field in msg_queue data structure provides list of messages in queue

Select one:

- ☐ 1. q_qbytes
- ☐ 2. q_messages
- ☐ 3. q_qnum
- ☒ 4. q_qbytes ✗

The correct answer is: q_messages

Question **116**

Correct

Mark 5.00 out of 5.00

.....function closes a POSIX message queue without destroying it

Select one:

- ☒ 1. mq_close() ✓
- ☐ 2. mq_unlink()
- ☐ 3. mq_notify()
- ☐ 4. mq_setattr()

The correct answer is: mq_close()

Question **117**

Correct

Mark 5.00 out of 5.00

No data blocks are required for one of these kinds of files.

Select one:

- ☒ 1. Named pipe ✓
- ☐ 2. Symbolic Link
- ☐ 3. Regular file
- ☐ 4. Directory

The correct answer is: Named pipe

Question **118**

Correct

Mark 5.00 out of 5.00

.....Best suited to implement producer/consumer interactions among processes.

Select one:

- ☒ 1. Pipe/FIFO ✓
- ☐ 2. Shared Memory
- ☐ 3. Message Queues
- ☐ 4. Semaphore

The correct answer is: Pipe/FIFO

Question **119**

Correct

Mark 10.00 out of 10.00

In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the Size of Inode Table in blocks in each blockgroup.

Select one:

- ☐ 1. 128 blocks
- ☐ 2. 256 blocks
- ☒ 3. 512 blocks ✓
- ☐ 4. 1024 blocks

The correct answer is: 512 blocks

Question **120**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides free blocks count in the filesystem.

Answer: ✓

The correct answer is: s_free_blocks_count

Question **121**

Correct

Mark 5.00 out of 5.00

.....function destroys a POSIX message queue

Select one:

- ☐ 1. mq_close()
- ☐ 2. mq_notify()
- ☐ 3. mq_open()
- ☒ 4. mq_unlink() ✓

The correct answer is: mq_unlink()

Question **122**

Correct

Mark 5.00 out of 5.00

The field of the inode, which is not used for regular files, represents a 32-bit extension of the i_size field.

Answer: ✓

The correct answer is: i_dir_acl

Question **123**

Correct

Mark 5.00 out of 5.00

In journaling mode, all filesystem data and metadata changes are logged into the journal.

Answer: ✓

The correct answer is: journal

Question **124**

Correct

Mark 5.00 out of 5.00

In Ext2 Block group descriptor structure, field provides Block number of block bitmap.

Answer: ✓

The correct answer is: bg_block_bitmap

Question **125**

Correct

Mark 5.00 out of 5.00

The size of each inode in the inode table in ext2 isbytes.

Select one:

- ☒ 1. 128 ✓
- ☐ 2. 32
- ☐ 3. 64
- ☐ 4. 256

The correct answer is: 128

Question **126**

Correct

Mark 5.00 out of 5.00

In Unix command shells, can be created by means of the | operator.

Select one:

- ☐ 1. Message Queues
- ☐ 2. Semaphore
- ☐ 3. FIFO
- ☒ 4. Pipe ✓

The correct answer is: Pipe

Question **127**

Correct

Mark 10.00 out of 10.00

In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 and 512 Blockgroups, then total number of blocks in a block group is blocks.

Select one:

- ☐ 1. 512
- ☐ 2. 2048
- ☒ 3. 8192 ✓
- ☐ 4. 1024

The correct answer is: 8192

Question **128**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"When creating an Ext2 filesystem, the system administrator may choose how many inodes to allow for a partition of a given size, depending on the expected number of files to be stored on it."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **129**

Correct

Mark 5.00 out of 5.00

Number of IPC semaphore resources that can be used by default is

Select one:

- ☐ 1. 32
- ☐ 2. 64
- ☒ 3. 128 ✓
- ☐ 4. 256

The correct answer is: 128

Question **130**

Correct

Mark 5.00 out of 5.00

Each IPC shared memory region is associated with a file belonging to thespecial filesystem located in VFS layer in Linux

Answer:



The correct answer is: shm

Question **131**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides total number of inodes in the filesystem

Select one:

- ☒ 1. s_inodes_count ✓
- ☐ 2. s_inode_size
- ☐ 3. s_inodes_per_group
- ☐ 4. s_free_inodes_count

The correct answer is: s_inodes_count

Question **132**

Correct

Mark 5.00 out of 5.00

.....field in msg_queue data structure provides number of bytes in queue

Select one:

- ☐ 1. q_qbytes
- ☒ 2. q_qcbytes ✓
- ☐ 3. q_messages
- ☐ 4. q_qnum

The correct answer is: q_qcbytes

Question **133**

Correct

Mark 5.00 out of 5.00

The function deletes a disk inode, which is identified by an inode object whose address inode is passed as the parameter.

Select one:

- ☐ 1. ext2_free_blocks()
- ☒ 2. ext2_free_inode() ✓
- ☐ 3. ext2_new_inode()
- ☐ 4. ext2_new_inode()

The correct answer is: ext2_free_inode()

Question **134**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"To minimize the impact of system crashes, when creating a new hard link for a file, the new name is added into the proper directory first and the counter of hard links in the disk inode is increased next."

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **135**

Correct

Mark 5.00 out of 5.00

In Unix System V, Size of each message in a queue by default is bytes

Answer: 8192 ✓

The correct answer is: 8192

Question **136**

Correct

Mark 5.00 out of 5.00

Number of pipes created by the shell when it executes the following command is

```
echo "WINE Is Not windows Emulator" | rev | cut -f1 -d' ' | rev | wc -c
```

Answer: 4 ✓

The correct answer is: 4

Question **137**

Correct

Mark 5.00 out of 5.00

System V IPC first appeared in a development Unix variant called

Select one:

- ☐ 1. Sun Solaris
- ☐ 2. AT&T's System III
- ☐ 3. Linux
- ☒ 4. Columbus Unix ✓

The correct answer is: Columbus Unix

Question **138**

Correct

Mark 10.00 out of 10.00

Given Ex2 filesystem with default data block size=4KB, write the i_size and i_blocks fields for a file hole when the following command is executed.

```
$ echo -n "X" | dd of=/tmp/hole bs=1024 seek=6
```

Select one:

- ☐ 1. i_size = 6145, i_blocks =1
- ☐ 2. i_size = 6144, i_blocks =8
- ☒ 3. i_size = 6145, i_blocks =8 ✓
- ☐ 4. i_size = 6144, i_blocks =1

The correct answer is: i_size = 6145, i_blocks =8

Question **139**

Correct

Mark 5.00 out of 5.00

Pipes are fully integrated into the layer, and the kernel can handle them in the same way as named pipes or FIFOs, which truly exist as files recognizable to end users.

Answer: VFS



The correct answer is: VFS

Question **140**

Correct

Mark 5.00 out of 5.00

How many processes are created by the shell when it executes the following command :

`$ ls | more`

Select one:

- ☐ 1. 3
- ☐ 2. 4
- ☒ 3. 2
- ☐ 4. 1



The correct answer is: 2

Question **141**

Correct

Mark 5.00 out of 5.00

..... function is invoked to create IPC Message Queue resource.

Select one:

- ☐ 1. semget()
- ☒ 2. msgget()
- ☐ 3. shmat()
- ☐ 4. shmget()



The correct answer is: msgget()

Question **142**

Correct

Mark 5.00 out of 5.00

Theincludes all sem_undo data structures corresponding to the processes that performed undoable operations on the semaphore.

Answer: per-semaphore list



The correct answer is: per-semaphore list

Question **143**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides number of blocks to preallocate for directories

Answer: s_prealloc_dir_blocks



The correct answer is: s_prealloc_dir_blocks

Question **144**

Correct

Mark 10.00 out of 10.00

Assume a data block size of 4KB, find the file block number for the given file offset 348320.

Select one:

- ☐ 1. 75
- ☒ 2. 85 ✓
- ☐ 3. 95
- ☐ 4. 105

The correct answer is: 85

Question **145**

Correct

Mark 5.00 out of 5.00

A file is a portion of a regular file that contains null characters and is not stored in any data block on disk.

Answer: ✓

The correct answer is: hole

Question **146**

Correct

Mark 5.00 out of 5.00

..... field in shmid_kernel data structure provides number of current attaches

Select one:

- ☐ 1. shm_segsz
- ☐ 2. shm_atim
- ☒ 3. shm_nattch ✓
- ☐ 4. shm_file

The correct answer is: shm_nattch

Question **147**

Correct

Mark 5.00 out of 5.00

sudo /etc is used to make a whole directory (e.g., /etc) including all its content immutable at once recursively,

Select one:

- ☐ 1. chattr +R +i
- ☐ 2. chattr +R -i
- ☒ 3. chattr -R +i ✓
- ☐ 4. chattr -R -i

The correct answer is: chattr -R +i

Question **148**

Correct

Mark 5.00 out of 5.00

..... sets Access Control Lists (ACLs) of files and directories.

Answer:



The correct answer is: setfacl

Question **149**

Correct

Mark 5.00 out of 5.00

Each IPC resource also has a-bit IPC identifier, which is somewhat similar to the file descriptor associated with an open file.

Select one:

- ☐ 1. 128
- ☐ 2. 64
- ☐ 3. 16
- ☒ 4. 32 ✓

The correct answer is: 32

Question **150**

Correct

Mark 5.00 out of 5.00

Select the basic mechanisms offered by Linux to allow inter-process communication:

Select one:

- ☐ 1. Pipes and FIFOs (named pipes)
- ☐ 2. Semaphores
- ☐ 3. Messages
- ☐ 4. Shared memory regions
- ☐ 5. Sockets
- ☒ 6. All of the above ✓

The correct answer is: All of the above

Question **151**

Correct

Mark 5.00 out of 5.00

..... function is invoked to create IPC Semaphore resource.

Select one:

- ☐ 1. shmat()
- ☐ 2. msgget()
- ☒ 3. semget() ✓
- ☐ 4. shmget()

The correct answer is: semget()

Question **152**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"When creating an Ext2 filesystem, the system administrator may choose the optimal block size (from 1,024 to 4,096 bytes), depending on the expected average file length".

Select one:

☒ True ✓

☐ False

The correct answer is 'True'.

Question **153**

Correct

Mark 5.00 out of 5.00

In Ext4, If the feature flag is set, redundant copies of the superblock and group descriptors are kept only in the groups whose group number is either 0 or a power of 3, 5, or 7.

Answer: ✓

The correct answer is: sparse_super

Question **154**

Correct

Mark 5.00 out of 5.00

Actually, in the 80 x 86 architecture, there is just one IPC system call named

Answer: ✓

The correct answer is: ipc()

Question **155**

Correct

Mark 10.00 out of 10.00

For given a 64GB partition having Ext2 file system with disk block size = 4KB, # of inodes = 8192, then the total number of blocks in a blockgroup is blocks.

Select one:

☐ 1. 2048

☒ 2. 32768 ✓

☐ 3. 16384

☐ 4. 65536

The correct answer is: 32768

Question **156**

Correct

Mark 5.00 out of 5.00

In ext2_inode structure, field provides Time of last file access.

Select one:

- ☐ 1. i_mtime
- ☒ 2. i_atime ✓
- ☐ 3. i_ctime
- ☐ 4. i_dtime

The correct answer is: i_atime

Question **157**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"When a new file is created, the values of the s_free_inodes_count field in the Ext2 superblock and of the bg_free_inodes_count field in the proper group descriptor must be decreased."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **158**

Correct

Mark 5.00 out of 5.00

A process invokes thewrapper function to test and decrease all primitive semaphore values involved.

Answer: ✓

The correct answer is: semop()

Question **159**

Correct

Mark 10.00 out of 10.00

In Ext4 file system, a 1,024-byte block contains inodes, while a 4,096-byte block contains inodes.

Select one:

- ☐ 1. 4, 32
- ☒ 2. 4, 16 ✓
- ☐ 3. 8, 32
- ☐ 4. 8, 64

The correct answer is: 4, 16

Question **160**

Correct

Mark 5.00 out of 5.00

The most important field in the sem_queue data structure for a pending request is, which is a pointer to an array of integer values describing each pending semaphore operation,

Select one:

- ☐ 1. status
- ☐ 2. sma
- ☐ 3. nsops
- ☒ 4. sops ✓

The correct answer is: sops

Question **161**

Correct

Mark 20.00 out of 20.00

Assume a data block size of 4KB, if the file block number is 1655 then compute the corresponding logical block number of the data block.

Select one:

- ☐ 1. i_block[12] & first level indirectblock[619]
- ☐ 2. i_block[14], first level indirectblock[1] & second level indirectblock0[72]
- ☒ 3. i_block[13], first level indirectblock[0] & second level indirectblock0[619] ✓
- ☐ 4. i_block[13], first level indirectblock[1] & second level indirectblock0[1655]

The correct answer is: i_block[13], first level indirectblock[0] & second level indirectblock0[619]

Question **162**

Incorrect

Mark 0.00 out of 5.00

Determine how many disk accesses (worst case) are required to access a file data block in 12th component i_block[12] of the i_block array.

Select one:

- ☐ 1. 3
- ☐ 2. 5
- ☒ 3. 4 ✗
- ☐ 4. 2

The correct answer is: 3

Question **163**

Correct

Mark 10.00 out of 10.00

For given a 64GB partition having Ext2 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the total number of blockgroups is blocks

Select one:

- ☐ 1. 2048
- ☐ 2. 1024
- ☐ 3. 256
- ☒ 4. 512 ✓

The correct answer is: 512

Question **164**

Incorrect

Mark 0.00 out of 5.00

Determine how many disk accesses (worst case) are required to access a file data block in one of the first 12 components i_block[0] to [11] of the i_block array.

Select one:

- ☐ 1. 2
- ☐ 2. 3
- ☒ 3. 4 ✗
- ☐ 4. 5

The correct answer is: 2

Question **165**

Incorrect

Mark 0.00 out of 10.00

In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the total number of blocks in each Blockgroup.

Select one:

- ☐ 1. 32767
- ☐ 2. 32768
- ☒ 3. 65535 ✗
- ☐ 4. 65536

The correct answer is: 65536

Question **166**

Correct

Mark 5.00 out of 5.00

" It is impossible for two arbitrary processes to share the same pipe, unless the pipe was created by a common ancestor process. "

Say True or false

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **167**

Correct

Mark 10.00 out of 10.00

..... field in superblock object provides List of file objects

Answer: ✓

The correct answer is: s_files

Question **168**

Correct

Mark 10.00 out of 10.00

In Ext4 file system, Journal length is

Select one:

- ☐ 1. 4096 blocks
- ☐ 2. 8192 blocks
- ☒ 3. 32768 blocks ✓
- ☐ 4. 16384 blocks

The correct answer is: 32768 blocks

Question **169**

Correct

Mark 5.00 out of 5.00

POSIX introduces a function namedspecifically to create a FIFO.

Answer: ✓

The correct answer is: mkfifo()

Question **170**

Correct

Mark 10.00 out of 10.00

In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 and 512 Blockgroups, then size of node table is blocks.

Select one:

- ☐ 1. 1024
- ☐ 2. 2048
- ☐ 3. 512
- ☒ 4. 256 ✓

The correct answer is: 256

Question **171**

Correct

Mark 10.00 out of 10.00

..... command with option is used to make a file immutable

Select one:

- ☒ 1. chattr +i ✓
- ☐ 2. chattr -i
- ☐ 3. chattr +a
- ☐ 4. chattr -a

The correct answer is: chattr +i

Question **172**

Correct

Mark 5.00 out of 5.00

The ext2_get_block() function handles the data structures and when necessary, invokes the function to actually search for a free block in the Ext2 partition.

Select one:

- ☐ 1. ext2_free_inode()
- ☐ 2. ex2_get_block()
- ☒ 3. ext2_alloc_block() ✓
- ☐ 4. ext2_new_inode()

The correct answer is: ext2_alloc_block()

Question **173**

Correct

Mark 5.00 out of 5.00

In a pipe_inode_info structure, field provides number of reading processes.

Answer: ✓

The correct answer is: readers

Question **174**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides free inodes count in the filesystem.

Answer: ✓

The correct answer is: s_free_inodes_count

Question **175**

Correct

Mark 5.00 out of 5.00

Number of processes created by the shell when it executes the following command is

\$ head -60 tfile | tail -1 | wc -c

Answer: ✓

The correct answer is: 3

Question **176**

Correct

Mark 10.00 out of 10.00

Determine how many disk accesses (worst case) are required to access a file data block in 14th component `i_block[14]` of the `i_block` array.

Select one:

- ☒ 1. 5 ✓
- ☐ 2. 6
- ☐ 3. 3
- ☐ 4. 4

The correct answer is: 5

Question **177**

Correct

Mark 10.00 out of 10.00

Consider an Ext2 1.44 MB floppy disk which is initialized by `make2fs` with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.

Compute the number of inodes in the inode table considering the exact number of blocks on the floppy.

Select one:

- ☒ 1. 184 ✓
- ☐ 2. 194
- ☐ 3. 174
- ☐ 4. 180

The correct answer is: 184

Question **178**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"File fragmentation increases the average time of sequential read operations on the files, because the disk heads must be frequently repositioned during the read operation."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **179**

Correct

Mark 5.00 out of 5.00

"The Ext2 filesystem status is stored in the `s_mount_state` field of the superblock on disk."

Say TRUE OR FALSE.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **180**

Correct

Mark 10.00 out of 10.00

Determine how many disk accesses (worst case) are required to access a file data block in 13th component `i_block[13]` of the `i_block` array

Select one:

- ☐ 1. 6
- ☐ 2. 7
- ☒ 3. 4 ✓
- ☐ 4. 5

The correct answer is: 4

Question **181**

Correct

Mark 5.00 out of 5.00

"Pages of an IPC shared memory region are swappable and not syncable because they map a special inode that has no image on disk."

Say True or False

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **182**

Correct

Mark 5.00 out of 5.00

Each is a set of one or more semaphore values, not just a single value like a kernel semaphore.

Answer: ✓

The correct answer is: IPC semaphore

Question **183**

Correct

Mark 10.00 out of 10.00

For a given 64-GB Ext4 partition with a 4-KB block size, the approximate number of Block Groups in the disk partition is

Select one:

- ☐ 1. 4096
- ☒ 2. 512 ✓
- ☐ 3. 1024
- ☐ 4. 256

The correct answer is: 512

Question **184**

Correct

Mark 5.00 out of 5.00

When shell executes the following command, process takes the input from first pipe and writes the output to the second pipe

```
$ head -60 tfile | tail -1 | wc -c
```

Select one:

- ☐ 1. shell
- ☐ 2. head -60 tfile
- ☒ 3. tail -1 ✓
- ☐ 4. wc -c

The correct answer is: tail -1

Question **185**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"In Ext2 filesystem, both the superblock and the group descriptors are always duplicated in each block group."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **186**

Correct

Mark 5.00 out of 5.00

In journaling mode, Ext3 filesystem groups metadata and relative data blocks so that data blocks are written to disk before the metadata.

Answer: ✓

The correct answer is: ordered

Question **187**

Correct

Mark 5.00 out of 5.00

....., Allow processes on different computers to exchange data through a network.

Select one:

- ☐ 1. Message Queues
- ☒ 2. Sockets ✓
- ☐ 3. Semaphore
- ☐ 4. Shared Memory

The correct answer is: Sockets

Question **188**

Correct

Mark 20.00 out of 20.00

Assume a data block size of 1KB, if the file block number is 340 then compute the corresponding logical block number of the data block.

Select one:

- ☒ 1. i_block[13], first level indirectblock[0], second level indirectblock0[72] ✓
- ☐ 2. i_block[13], first level indirectblock[0] & second level indirectblock0[85]
- ☐ 3. i_block[12] & first level indirectblock[72]
- ☐ 4. i_block[14], first level indirectblock[1] & second level indirectblock0[72]

The correct answer is: i_block[13], first level indirectblock[0], second level indirectblock0[72]

Question **189**

Correct

Mark 5.00 out of 5.00

One pipe buffer consists of page frame(s)

Select one:

- ☐ 1. 16
- ☐ 2. 8
- ☒ 3. 1 ✓
- ☐ 4. 4

The correct answer is: 1

Question **190**

Correct

Mark 5.00 out of 5.00

.....function establishes an asynchronous notification mechanism for the arrival of messages in an empty POSIX message queue

Select one:

- ☐ 1. mq_open()
- ☒ 2. mq_notify() ✓
- ☐ 3. mq_unlink()
- ☐ 4. mq_close()

The correct answer is: mq_notify()

Question **191**

Correct

Mark 5.00 out of 5.00

In ext2_inode structure, field provides file length in bytes

Select one:

- ☐ 1. i_block
- ☐ 2. i_blocks
- ☒ 3. i_size ✓
- ☐ 4. i_file_acl

The correct answer is: i_size

Question **192**

Correct

Mark 5.00 out of 5.00

In Ext2 superbblock structure, field provides time of last write operation.

Select one:

- ☐ 1. s_mnttime
- ☐ 2. s_checkinterval
- ☒ 3. s_wtime ✓
- ☐ 4. s_mtime

The correct answer is: s_wtime

Question **193**

Correct

Mark 5.00 out of 5.00

The number of blocks required to store Data Block Bitmap in a Block Group is

Answer: ✓

The correct answer is: 1

Question **194**

Correct

Mark 5.00 out of 5.00

..... is a special filesystem used in linux which is located in VFS layer in memory to expedite the handling of pipes.

Answer: ✓

The correct answer is: pipefs

Question **195**

Correct

Mark 5.00 out of 5.00

No data blocks are required for one of these kinds of files.

Select one:

- ☐ 1. Regular file
- ☐ 2. Directory
- ☒ 3. Socket ✓
- ☐ 4. Symbolic Link

The correct answer is: Socket

Question **196**

Correct

Mark 10.00 out of 10.00

For given a 64GB partition having Ext4 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the number of blocks required to store Inode table in a block group is blocks

Select one:

- ☐ 1. 256
- ☒ 2. 512 ✓
- ☐ 3. 128
- ☐ 4. 1024

The correct answer is: 512

Question **197**

Correct

Mark 5.00 out of 5.00

..... inode number in a directory entry of a directory file indicates that the file specified in that entry has been deleted.

Answer:

The correct answer is: 0

Question **198**

Correct

Mark 5.00 out of 5.00

The function creates an Ext2 disk inode, returning the address of the corresponding inode object.

Select one:

- ☒ 1. ext2_new_inode() ✓
- ☐ 2. ex2_get_block()
- ☐ 3. ext2_free_inode()
- ☐ 4. ext2_free_blocks()

The correct answer is: ext2_new_inode()

Question **199**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides time of last mount operation.

Select one:

- ☒ 1. s_mtime ✓
- ☐ 2. s_checkinterval
- ☐ 3. s_wtime
- ☐ 4. s_mnttime

The correct answer is: s_mtime

Question **200**

Correct

Mark 10.00 out of 10.00

Suppose that each block group contains 4,096 inodes, the address on disk of inode 13,021 can be given as

Select one:

- ☐ 1. Blockgroup 0, 13021
- ☒ 2. Blockgroup 3, 733 ✓
- ☐ 3. Blockgroup 3, 13021
- ☐ 4. Blockgroup 2, 733

The correct answer is: Blockgroup 3, 733

Question **201**

Correct

Mark 5.00 out of 5.00

In Ext4 file system, Flex Block group size (number of Blockgroups combined in a flex Block group) is

Answer:

16 ✓

The correct answer is: 16

Question **202**

Correct

Mark 5.00 out of 5.00

In ext2_inode structure, field provides Time that inode last changed.

Select one:

- ☐ 1. i_mtime
- ☒ 2. i_ctime ✓
- ☐ 3. i_atime
- ☐ 4. i_dtime

The correct answer is: i_ctime

Question **203**

Correct

Mark 10.00 out of 10.00

For given a 64GB partition having Ext4 file system with disk block size = 8KB, # of inodes = 8192 inodes, then the total number of blocks in a blockgroup is blocks

Select one:

- ☐ 1. 32768
- ☐ 2. 16384
- ☒ 3. 65536 ✓
- ☐ 4. 2048

The correct answer is: 65536

Question **204**

Correct

Mark 10.00 out of 10.00

If disk block size = 32KB, # of inodes = 32768 and Partition size = 512GB then determine the Size of one blockgroup in the file system.

Select one:

- ☒ 1. 8GB ✓
- ☐ 2. 512MB
- ☐ 3. 4GB
- ☐ 4. 2GB

The correct answer is: 8GB

Question **205**

Correct

Mark 5.00 out of 5.00

The Regular file is associated with a file type number

Answer:

1 ✓

The correct answer is: 1

Question **206**

Correct

Mark 10.00 out of 10.00

Assume a data block size of 1KB, if the file block number is 13 then compute the corresponding logical block number of the data block.

Select one:

- ☒ 1. i_block[12] & first level indirectblock[1] ✓
- ☐ 2. i_block[11]
- ☐ 3. i_block[13] & first level indirectblock[3]
- ☐ 4. i_block[12] & first level indirectblock[2]

The correct answer is: i_block[12] & first level indirectblock[1]

Question **207**

Correct

Mark 5.00 out of 5.00

A is a one-way flow of data between processes.

Select one:

- ☐ 1. Message Queues
- ☐ 2. FIFO
- ☒ 3. Pipe ✓
- ☐ 4. Semaphore

The correct answer is: Pipe

Question **208**

Correct

Mark 5.00 out of 5.00

The Socket file is associated with a file type number

Answer: 6



The correct answer is: 6

Question **209**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"In Ext4 filesystem, both the superblock and the group descriptors are always duplicated in each block group."

Select one:

☐ True

☒ False



The correct answer is 'False'.

Question **210**

Correct

Mark 5.00 out of 5.00

..... function creates a POSIX message queue

Select one:

☐ 1. mq_unlink()

☐ 2. mq_notify()

☒ 3. mq_open()



☐ 4. mq_close()

The correct answer is: mq_open()

Question **211**

Correct

Mark 5.00 out of 5.00

Number of primitive semaphores available inside a single IPC semaphore resource by default is

Select one:

☐ 1. 30

☒ 2. 250



☐ 3. 256

☐ 4. 128

The correct answer is: 250

Question **212**

Correct

Mark 10.00 out of 10.00

In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 and Partition size = 256GB then determine the Total number of blockgroups.

Select one:

- ☐ 1. 8192
- ☒ 2. 512 ✓
- ☐ 3. 32786
- ☐ 4. 4096

The correct answer is: 512

Question **213**

Correct

Mark 20.00 out of 20.00

Assume a data block size of 1KB, if the file block number is 340 then compute the corresponding logical block number of the data block.

Select one:

- ☐ 1. i_block[12] & first level indirectblock[72]
- ☐ 2. i_block[14], first level indirectblock[1] & second level indirectblock0[72]
- ☒ 3. i_block[13], first level indirectblock[0], second level indirectblock0[72] ✓
- ☐ 4. i_block[13], first level indirectblock[0] & second level indirectblock0[85]

The correct answer is: i_block[13], first level indirectblock[0], second level indirectblock0[72]

Question **214**

Correct

Mark 10.00 out of 10.00

Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.

Compute the size of inode table in terms of blocks

Select one:

- ☐ 1. 30
- ☒ 2. 23 ✓
- ☐ 3. 20
- ☐ 4. 33

The correct answer is: 23

Question **215**

Correct

Mark 10.00 out of 10.00

In Ext2 file system, a 1,024-byte block contains inodes, while a 4,096-byte block contains inodes.

Select one:

- ☒ 1. 8, 32 ✓
- ☐ 2. 4, 16
- ☐ 3. 8, 64
- ☐ 4. 4, 32

The correct answer is: 8, 32

Question **216**

Correct

Mark 5.00 out of 5.00

A process creates a new pipe by means of thesystem call.

Answer: ✓

The correct answer is: pipe()

Question **217**

Correct

Mark 5.00 out of 5.00

When a process chooses to use fail-safe mechanism, the resulting operations are calledsemaphore operations.

Select one:

- ☐ 1. kernel
- ☒ 2. undoable ✓
- ☐ 3. primitive
- ☐ 4. doable

The correct answer is: undoable

Question **218**

Correct

Mark 15.00 out of 15.00

Consider Data Block addressing in Ext2/ext4 disk file system with a block size of 4KB.

Determine the maximum size of the file that can be addressed by 13th component of the i_block array.

Select one:

- ☐ 1. 4TB
- ☐ 2. 4MB
- ☐ 3. 48KB
- ☒ 4. 4GB ✓

The correct answer is: 4GB

Question **219**

Correct

Mark 20.00 out of 20.00

For a given 64GB partition having Ext2 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the number of data blocks available for storing file data is blocks

Select one:

- ☐ 1. 32768
- ☒ 2. 32506 ✓
- ☐ 3. 259
- ☐ 4. 771

The correct answer is: 32506

Question **220**

Correct

Mark 5.00 out of 5.00

In ext2_inode structure, field provides Time that file contents last changed

Select one:

- ☒ 1. i_mtime ✓
- ☐ 2. i_dtime
- ☐ 3. i_ctime
- ☐ 4. i_atime

The correct answer is: i_mtime

Question **221**

Correct

Mark 5.00 out of 5.00

..... field in shmid_kernel data structure provides PID of last accessing process

Select one:

- ☐ 1. shm_atim
- ☐ 2. shm_nattch
- ☐ 3. shm_cprid
- ☒ 4. shm_lprid ✓

The correct answer is: shm_lprid

Question **222**

Correct

Mark 5.00 out of 5.00

In ext2_inode structure, field provides Time of file deletion.

Select one:

- ☐ 1. i_atime
- ☐ 2. i_mtime
- ☒ 3. i_dtime ✓
- ☐ 4. i_ctime

The correct answer is: i_dtime

Question **223**

Correct

Mark 10.00 out of 10.00

If disk block size = 16KB, # of inodes = 32768 inodes and Partition size = 512GB then determine the Number of blockgroups in the file system

Select one:

- ☐ 1. 1024
- ☒ 2. 256 ✓
- ☐ 3. 512
- ☐ 4. 128

The correct answer is: 256

Question **224**

Correct

Mark 5.00 out of 5.00

Full form of Ext2 is

Answer: ✓

The correct answer is: Second Extended Filesystem

Question **225**

Correct

Mark 10.00 out of 10.00

For a given 64-GB Ext4 partition with an 8-KB block size, the approximate number of Block Groups in the disk partition is

Select one:

- ☒ 1. 128 ✓
- ☐ 2. 256
- ☐ 3. 4096
- ☐ 4. 1024

The correct answer is: 128

Question **226**

Correct

Mark 5.00 out of 5.00

In Ext4 filesystem, when sb,s_log_block_size =2, Block size =

Select one:

- ☐ 1. 1KB
- ☐ 2. 64KB
- ☐ 3. 8KB
- ☒ 4. 4KB ✓

The correct answer is: 4KB

Question **227**

Correct

Mark 5.00 out of 5.00

The Block device file is associated with a file type number

Answer: ✓

The correct answer is: 4

Question **228**

Correct

Mark 5.00 out of 5.00

..... is a data structure in a Unix-style file system which describes a filesystem object such as a file or a directory.

Answer: ✓

The correct answer is: inode

Question **229**

Correct

Mark 10.00 out of 10.00

Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.

Compute the Total number of reserved blocks considering the exact number of blocks on the floppy.

Select one:

- ☐ 1. 80
- ☐ 2. 83
- ☐ 3. 70
- ☒ 4. 73 ✓

The correct answer is: 73

Question **230**

Correct

Mark 5.00 out of 5.00

A process creates a FIFO by issuing asystem call passing to it as parameters the pathname of the new FIFO and the value S_IFIFO (0x10000) logically ORed with the permission bit mask of the new file.

Answer: ✓

The correct answer is: mknod()

Question **231**

Correct

Mark 5.00 out of 5.00

Each pipe makes use of pipe buffers which greatly enhances the performance of User Mode applications that write large chunks of data in a pipe.

Select one:

- ☐ 1. 4
- ☒ 2. 16 ✓
- ☐ 3. 8
- ☐ 4. 1

The correct answer is: 16

Question **232**

Correct

Mark 15.00 out of 15.00

Assume a data block size of 4KB, if the file block number is 85 then compute the corresponding logical block number of the data block.

Select one:

- ☐ 1. i_block[13], first level indirectblock[73] & second level indirectblock0[85]
- ☐ 2. i_block[12] & first level indirectblock[85]
- ☒ 3. i_block[12] & first level indirectblock[73] ✓
- ☐ 4. i_block[13], first level indirectblock[85] & second level indirectblock1[73]

The correct answer is: i_block[12] & first level indirectblock[73]

Question **233**

Correct

Mark 10.00 out of 10.00

..... command with option is used to make a file append-only.

Select one:

- ☐ 1. chattr +i
- ☒ 2. chattr +a ✓
- ☐ 3. chattr -i
- ☐ 4. chattr -a

The correct answer is: chattr +a

Question **234**

Correct

Mark 5.00 out of 5.00

In a pipe_inode_info structure, field provides Read position in current pipe buffer.

Answer:

The correct answer is: start

Question **235**

Correct

Mark 5.00 out of 5.00

If the pathname of a symbolic link has up to characters, it is stored in the i_block field of the inode.

Answer: ✓

The correct answer is: 60

Question **236**

Correct

Mark 10.00 out of 10.00

Assume a data block size of 1KB, find the file block number for the given file offset 348320.

Select one:

- ☐ 1. 540
- ☐ 2. 240
- ☐ 3. 440
- ☒ 4. 340 ✓

The correct answer is: 340

Question **237**

Correct

Mark 10.00 out of 10.00

Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.

Compute the Number of Block group descriptors.

Select one:

- ☐ 1. 20
- ☒ 2. 1 ✓
- ☐ 3. 30
- ☐ 4. 2

The correct answer is: 1

Question **238**

Correct

Mark 5.00 out of 5.00

A file's inode number can be found using the command.

Answer: ✓

The correct answer is: ls -i

Question **239**

Correct

Mark 10.00 out of 10.00

The Ext3 journal is usually stored in a hidden file named located in the root directory of the filesystem.

Answer:



The correct answer is: .journal

Question **240**

Correct

Mark 5.00 out of 5.00

If the pathname of a symbolic link is longer than characters, a single data block is required

Answer:



The correct answer is: 60

Question **241**

Correct

Mark 10.00 out of 10.00

In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 inodes and 512 Blockgroups, then total number of disk blocks required to store Block Group descriptors in a block group is blocks

Select one:

- ☒ 1. 12 ✓
- ☐ 2. 6
- ☐ 3. 3
- ☐ 4. 24

The correct answer is: 12

Question **242**

Correct

Mark 10.00 out of 10.00

For given a 64GB partition having Ext4 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the total number of blockgroups is blocks

Select one:

- ☒ 1. 512 ✓
- ☐ 2. 256
- ☐ 3. 1024
- ☐ 4. 2048

The correct answer is: 512

Question **243**

Correct

Mark 5.00 out of 5.00

To avoid race conditions on the pipe's data structures, the kernel makes use of the semaphore included in the inode object.

Answer:



The correct answer is: i_sem

Question **244**

Correct

Mark 5.00 out of 5.00

In Ext2 superblock structure, field provides Block group number of this superblock.

Select one:

- ☒ 1. s_block_group_nr ✓
- ☐ 2. s_r_blocks_count
- ☐ 3. s_blocks_per_group
- ☐ 4. s_blocks_count

The correct answer is: s_block_group_nr

Question **245**

Correct

Mark 5.00 out of 5.00

The following command displays the blocks which are reserved as bad in the filesystem.

Select one:

- ☐ 1. \$dumpe2fs -f /dev/sda1
- ☐ 2. \$dumpe2fs -h /dev/sda1
- ☒ 3. \$dumpe2fs -b /dev/sda1 ✓
- ☐ 4. \$dumpe2fs -x /dev/sda1

The correct answer is: \$dumpe2fs -b /dev/sda1

Question **246**

Correct

Mark 5.00 out of 5.00

The Character device file is associated with a file type number

Answer: 3

The correct answer is: 3

Question **247**

Correct

Mark 5.00 out of 5.00

SAY TRUE OR FALSE

"If the kernel appends some data to an existing file so that the number of data blocks allocated for it increases, the values of the s_free_blocks_count field in the Ext2 superblock and of the bg_free_blocks_count field in the group descriptor must be modified."

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **248**

Correct

Mark 5.00 out of 5.00

The Named pipe (FIFO) file is associated with a file type number

Answer: ✓

The correct answer is: 5

Question **249**

Correct

Mark 10.00 out of 10.00

For each pipe, the kernel creates inode object plus file objects

Select one:

- ☐ 1. 2, 4
- ☐ 2. 2, 2
- ☒ 3. 1, 2 ✓
- ☐ 4. 1, 4

The correct answer is: 1, 2

Question **250**

Correct

Mark 5.00 out of 5.00

The number of Reserved GDT blocks in Ext4 Filesystem is

Select one:

- ☐ 1. 512
- ☒ 2. 1021 ✓
- ☐ 3. 128
- ☐ 4. 1024

The correct answer is: 1021

Question **251**

Correct

Mark 5.00 out of 5.00

In a pipe_inode_info structure, field provides number of writing processes.

Answer: ✓

The correct answer is: writers

Question **252**

Correct

Mark 5.00 out of 5.00

"In fact, metadata's log records are sufficient to restore the consistency of the on-disk filesystem data structures."

SAY TRUE OR FALSE

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **253**

Correct

Mark 10.00 out of 10.00

.....fields in the Ext2 file system data structures are always cached.

Select one or more:

- ☐ 1. Inode Bitmap
- ☐ 2. Inode Table
- ☐ 3. Block Bitmap
- ☒ 4. Superblock ✓
- ☒ 5. Block Group Descriptors ✓

The correct answers are: Superblock, Block Group Descriptors